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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

Student Name: Durva Kadam Student Roll No: 23

Experiment – 5 : Storage as a Service

Aim: To study and Implement Storage as a Service using different cloud platform AWS / GC / AZURE (Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to –

- Perform Amazon S3 Operations (Create/delete a bucket, Write an object, Read an object, Delete an object)
- Bucket versioning
- Know how to enable static website hosting on Amazon S3
- Know how to use lifecycle configuration rules

Lab objective mapped : ITDO802.3 : To learn, simulate and examine cloud service model and deploy IAAS, SAAS, PAAS services over different cloud platforms.

Prerequisite: Concept of Operating System, Infrastructure.

Requirements: Cloud Login**,** Desktop, Browser, Internet etc.

Pre-Experiment Theory:

CLOUD Storage services (describe-soft copy form)

S3 (describe-soft copy form)

Azure Blob Storage (describe-soft copy form)

Procedure
Create an Amazon Simple Storage Service (Amazon S3) Bucket in your selected region.

1. Log in to the AWS Management Console.

2. Choose an appropriate region.

3. Navigate to the Amazon S3 console. Notice that the region indicator now says

Global.

4. Start the create bucket process.

5. When prompted for Bucket Name, use unique name.

6. Choose a region.

7. Try to create the bucket.

8.You should now have a new Amazon S3 bucket.

Upload, Make Public, Rename, and Delete Objects in Your Bucket

Upload an Object

1. Load your new bucket in the Amazon S3 console.

2. Select Upload, then Add Files.

3. Locate a file on your PC.

4. Select a suitable file, then Start Upload. You will see the status of your file in the Transfers section.

5. After your file is uploaded, the status should change to Done.

Open the Amazon S3 URL

6. Now open the properties for the object. The properties should include bucket, name, and link.

7. Copy the Amazon S3 URL for the object.

8. Paste the URL in the address bar of a new browser window or tab. You should get a message with an XML error code AccessDenied. Even though the object has a URL, it is private by default, so it cannot be accessed by a web browser.

Make the Object Public

9. Go back to the Amazon S3 Console and select Make Public.

10. Copy the Amazon S3 URL again and try to open it in a browser or tab. Your public image file should now display in the browser or browser tab.

Rename Object

11. In the Amazon S3 console, select Rename.

12. Rename the object, but keep the same file extension.

13. Copy the new Amazon S3 URL and try to open it in a browser or tab. You should see

the same image file.

Delete the Object

14. In the Amazon S3 console, select Delete. Select OK when prompted if you want to delete the object.

15. The object has now been deleted.

16. To verify, try to reload the deleted object's Amazon S3 URL. You should once again get the XML AccessDenied error message.

Enable Version Control

1. In the Amazon S3 console, load the properties of your bucket. Don't open the bucket.

2. Enable versioning in the properties and select OK to verify. Your bucket now has versioning enabled.

<u>Create Multiple Versions of an Object</u>

3. Create a text file named foo.txt on your computer and write your name in the text file.

4. Save the text file to a location of your choosing.

5. Upload the text file to your bucket. This will be version 1.

6. After you have uploaded the text file to your bucket, open the copy on your local computer and change your name to your surname. Save the text file with the original filename.

7. Upload the modified file to your bucket.

8. Select Show Versions on the uploaded object.

Post-Experiments Exercise

Enable Static Hosting on Your Bucket

1. Select your bucket in the Amazon S3 console.

2. In the Properties section, select Enable Website Hosting.

3. For the index document name, enter index.txt, and for the error document name, enter error.txt.

4. Use a text editor to create two text files and save them as index.txt and error.txt. In the index.txt file, write the phrase "Hello World," and in the error.txt file, write the phrase "Error Page." Save both text files and upload them to your bucket.

5. Make the two objects public.

6. Copy the Endpoint: link under Static Website Hosting and paste it in a browser window or tab. You should now see the phrase "Hello World" displayed.

7. In the address bar in your browser, try adding a forward slash followed by a made-up filename (for example, /test.html). You should now see the phrase "Error Page" displayed.

8. To clean up, delete all of the objects in your bucket and then delete the bucket itself.

Extended Theory:

1. What is storage life cycle and its significance? (to be written in hand)
2. S3 bucket pricing model with example (to be written in hand)
3. Block level storage vs Object level storage(soft copy)

Results/Calculations/Observations:

Fill the following observation tables-(to be written in hand)

Sr. No		
1.	In what ways does Amazon Simple Storage Service (Amazon S3) object storage differ from block and file storage? (Choose 2 answers)	A. Amazon S3 stores data in fixed size blocks. B. Objects are identified by a numbered address. C. Objects can be any size. D. Objects contain both data and metadata. E. Objects are stored in buckets.
2.	Name one more appropriates use case for Amazon Simple Storage Service (Amazon S3)?	1. Primary storage for a database 2. _____
3.	Your application stores critical data in Amazon Simple Storage Service, which must be protected against inadvertent or intentional deletion. Discuss two solutions?	1. _____ 2. _____

Post Experimental Exercise-

Questions:

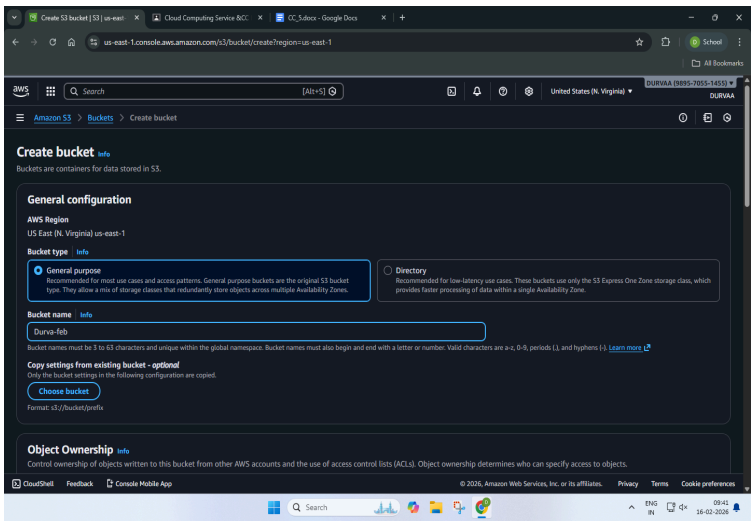
1. A certificate course on GCP (1Hr)
https://www.mygreatlearning.com/academy/learn-for-free/courses/google-cloud-platform-for-beginners1?utm_source=chatbot&utm_medium=new_flow&utm_campaign=job
Submit test score and certificate of the attended course

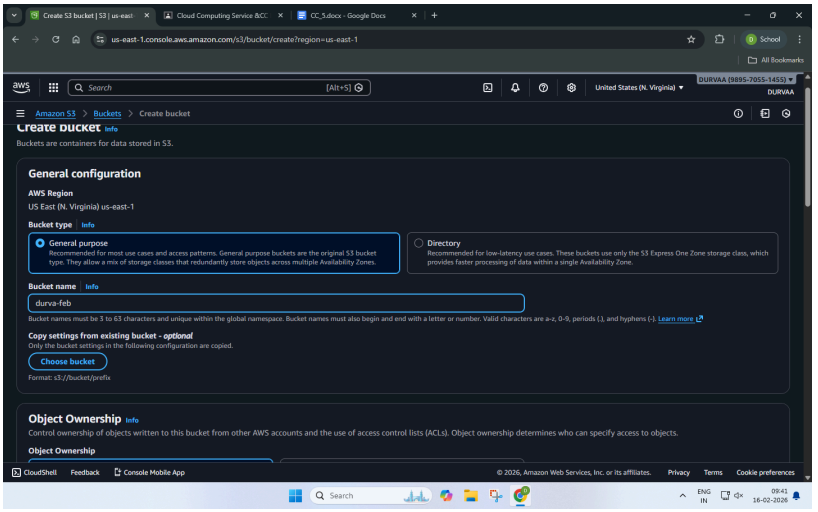
Conclusion:

1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:

[1] [Online]https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/concepts.html
[2] [Online]https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary


Create S3 bucket | S3 | us-east-1
Cloud Computing Service &CC
CC_5.docx - Google Docs
us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1
Amazon S3 > Buckets > Create bucket
Create bucket Info
Buckets are containers for data stored in S3.
General configuration
AWS Region
US East (N. Virginia) us-east-1
Bucket type Info
General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
Directory
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.
Bucket name Info
Durva-feb
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). Learn more
Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.
Choose bucket
Format: s3://bucket/prefix
Object Ownership Info
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.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Cloud Computing Service &CC
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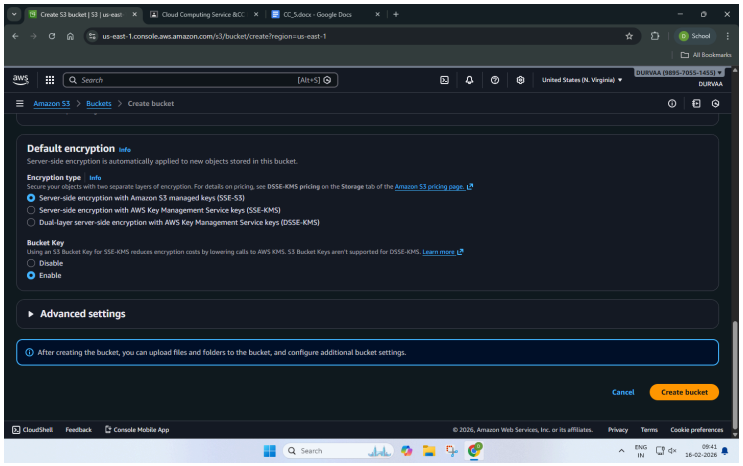
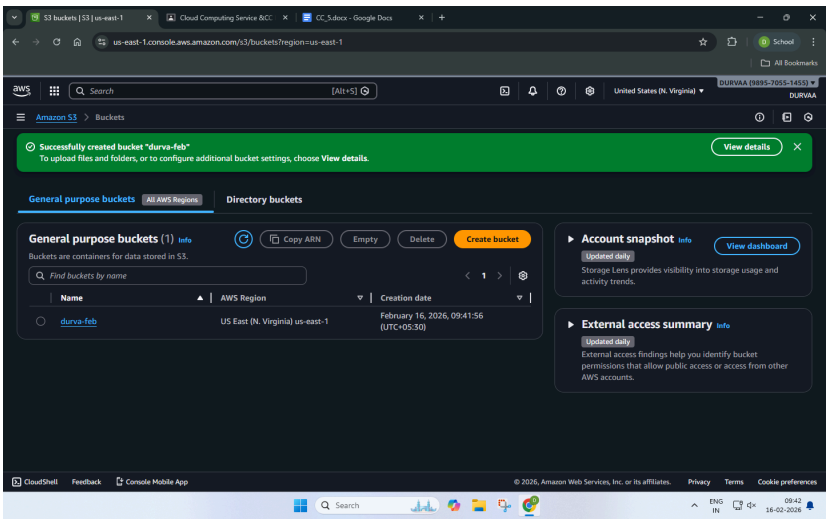
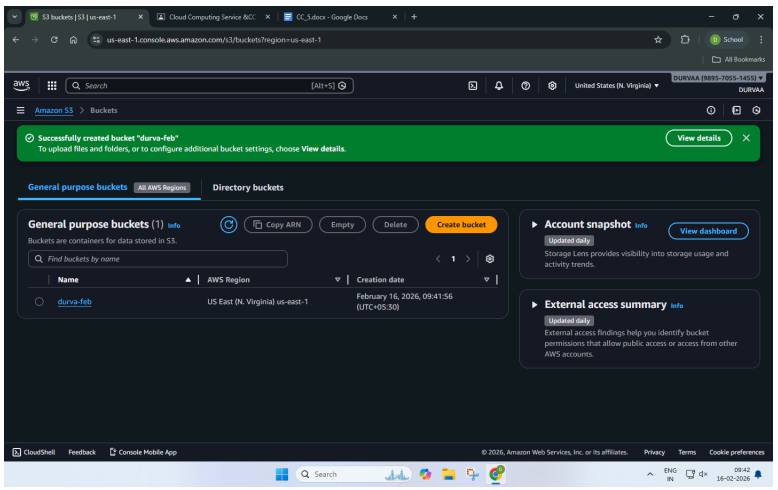

Create S3 bucket | S3 | us-east-1
Cloud Computing Service &CC
CC_5.docx - Google Docs
us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1
Amazon S3 > Buckets > Create bucket
Default encryption Info
Server-side encryption is automatically applied to new objects stored in this bucket.
Encryption type Info
Secure your objects with two separate layers of encryption. For details on pricing, see DSSE-KMS pricing on the Storage tab of the Amazon S3 pricing page.
Server-side encryption with Amazon S3 managed keys (SSE-S3)
Server-side encryption with AWS Key Management Service keys (SSE-KMS)
Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. Learn more
Disable
Enable
Advanced settings
After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.
Cancel
Create bucket


S3 buckets | S3 | us-east-1
Cloud Computing Service &CC
CC_5.docx - Google Docs
us-east-1.console.aws.amazon.com/s3/buckets?region=us-east-1
Amazon S3 > Buckets
Successfully created bucket "durva-feb"
To upload files and folders, or to configure additional bucket settings, choose View details.
View details
General purpose buckets
All AWS Regions
Directory buckets
General purpose buckets (1) Info
Buckets are containers for data stored in S3.
Copy ARN
Empty
Delete
Create bucket
Find buckets by name
Name
AWS Region
Creation date
durva-feb
US East (N. Virginia) us-east-1
February 16, 2026, 09:41:56 (UTC+05:30)
Account snapshot Info
View dashboard
Updated daily
Storage Lens provides visibility into storage usage and activity trends.
External access summary Info
Updated daily
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.


S3 buckets | S3 | us-east-1
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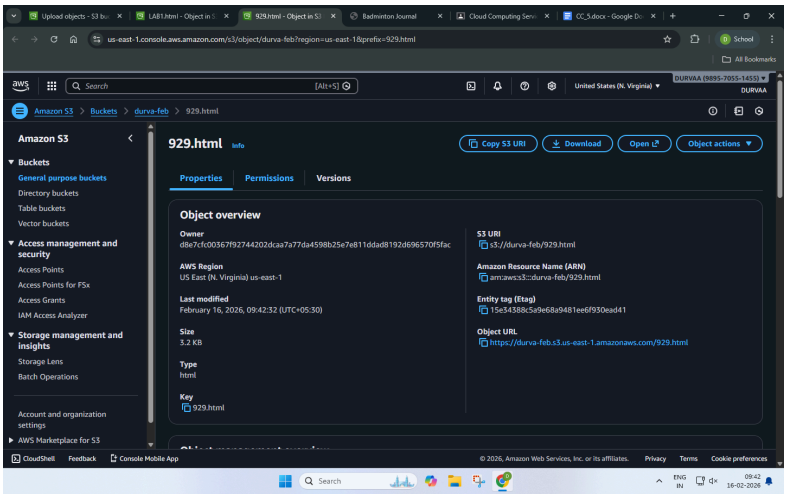
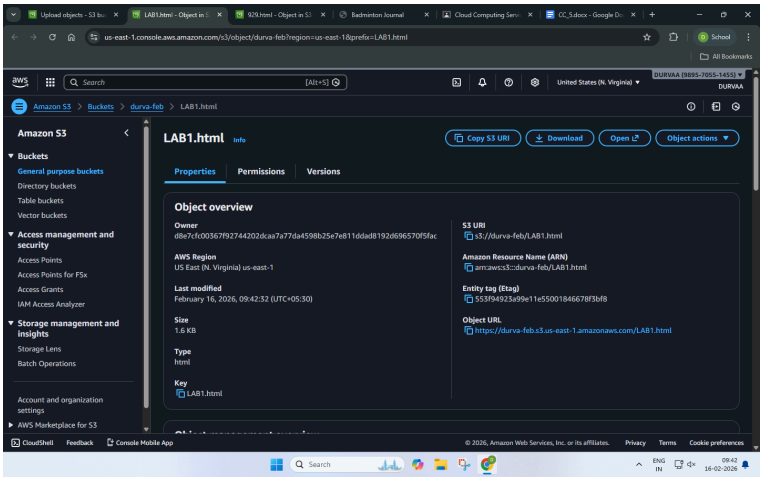
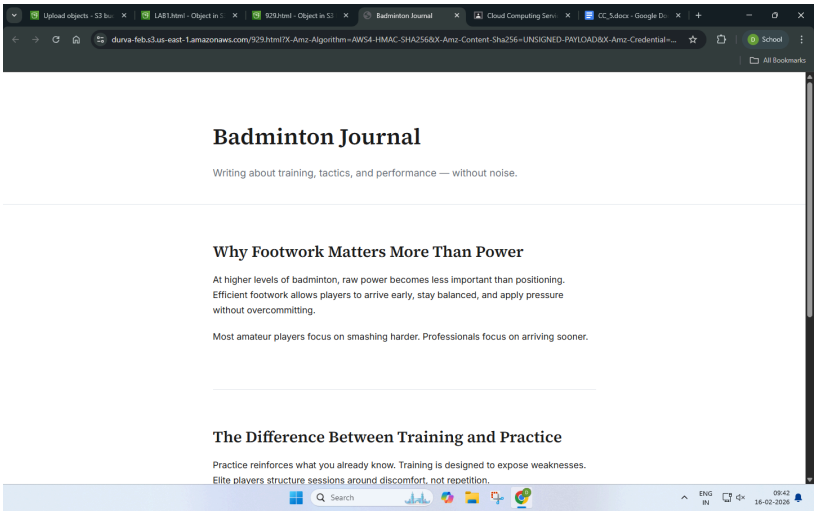

us-east-1.console.aws.amazon.com/s3/object/durva-feb?region=us-east-1&prefix=929.html
Amazon S3 > Buckets > durva-feb > 929.html
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
Account and organization settings
AWS Marketplace for S3
929.html Info
Copy S3 URI
Download
Open
Object actions
Properties
Permissions
Versions
Object overview
Owner
d8e7cfc00367f92744202dcaa7a77da4598b25e7e811ddad8192d696570f5fac
AWS Region
US East (N. Virginia) us-east-1
Last modified
February 16, 2026, 09:42:32 (UTC+05:30)
Size
3.2 KB
Type
html
Key
929.html
S3 URI
s3://durva-feb/929.html
Amazon Resource Name (ARN)
arn:aws:s3:::durva-feb/929.html
Entity tag (Etag)
15e34388c5a9e68a9481ee6f930ead41
Object URL
https://durva-feb.s3.us-east-1.amazonaws.com/929.html


us-east-1.console.aws.amazon.com/s3/object/durva-feb?region=us-east-1&prefix=LAB1.html
Amazon S3 > Buckets > durva-feb > LAB1.html
LAB1.html Info
Copy S3 URI
Download
Open
Object actions
Properties
Permissions
Versions
Object overview
Owner
d8e7cfc00367f92744202dcaa7a77da4598b25e7e811ddad8192d696570f5fac
AWS Region
US East (N. Virginia) us-east-1
Last modified
February 16, 2026, 09:42:32 (UTC+05:30)
Size
1.6 KB
Type
html
Key
LAB1.html
S3 URI
s3://durva-feb/LAB1.html
Amazon Resource Name (ARN)
arn:aws:s3:::durva-feb/LAB1.html
Entity tag (Etag)
553f94923a99e11e55001846678f3bf8
Object URL
https://durva-feb.s3.us-east-1.amazonaws.com/LAB1.html


durva-feb.s3.us-east-1.amazonaws.com/929.html?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Content-Sha256=UNSIGNED-PAYLOAD&X-Amz-Credential=...
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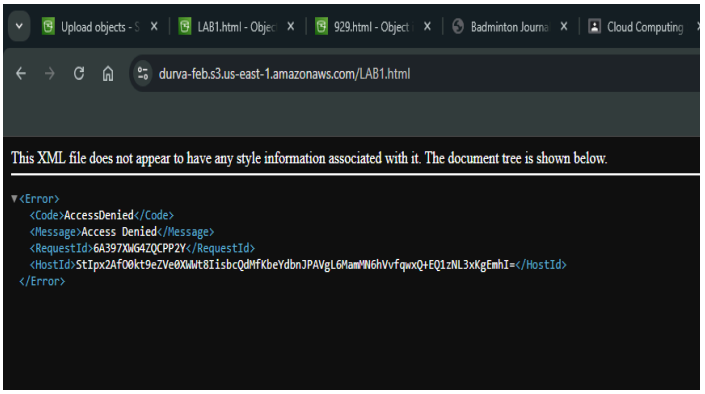
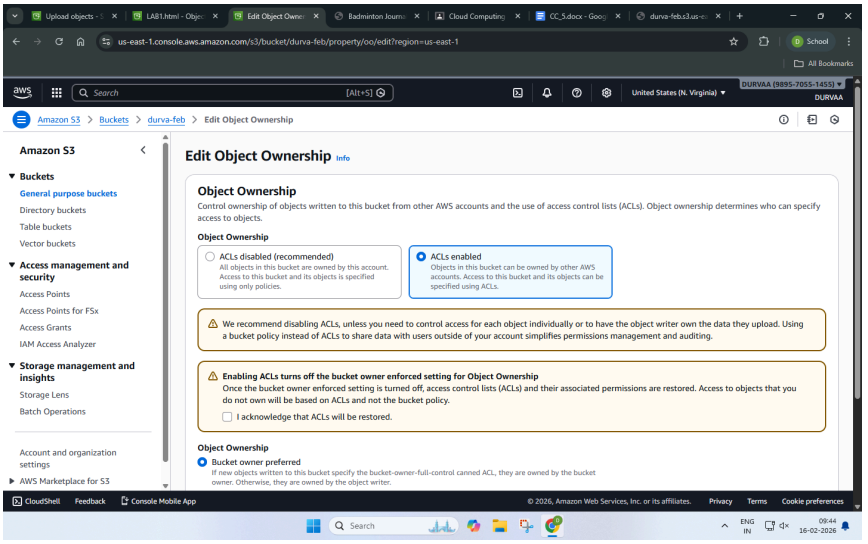
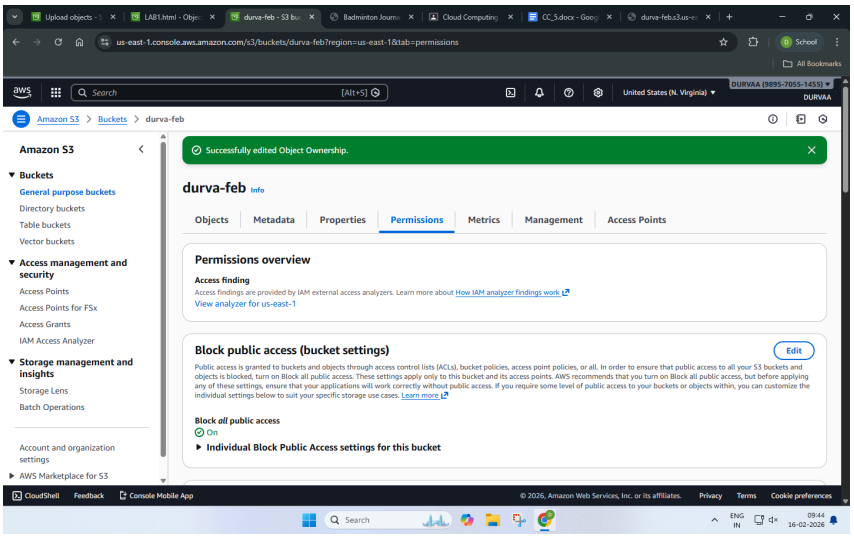
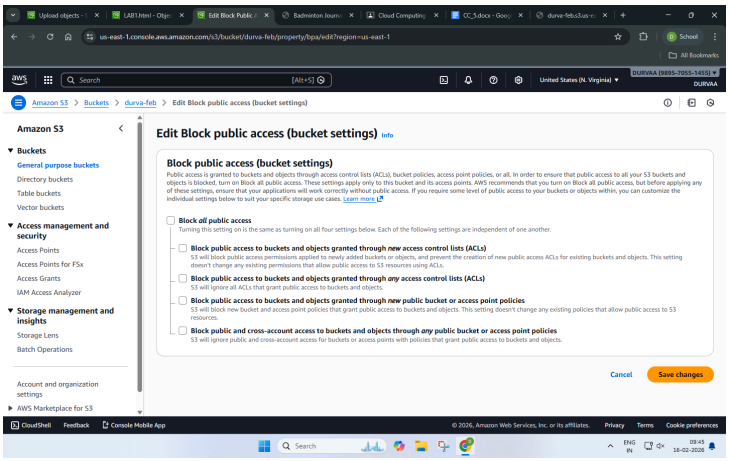
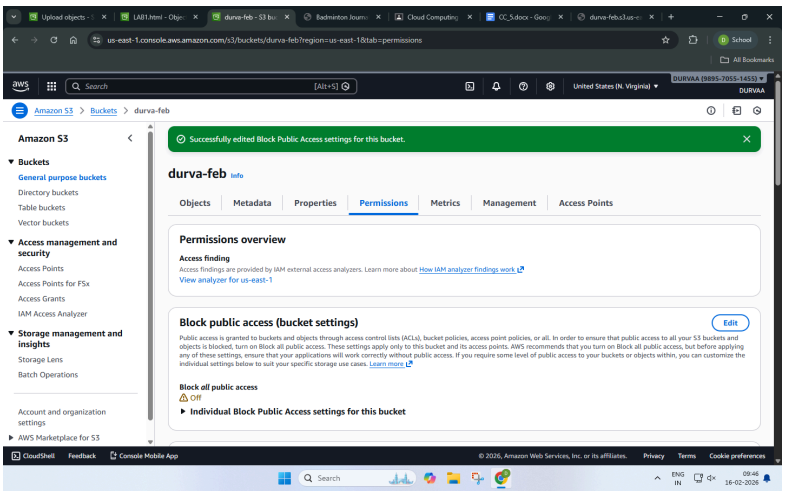

durva-feb.s3.us-east-1.amazonaws.com/LAB1.html
This XML file does not appear to have any style information associated with it. The document tree is shown below.
<Error>
<Code>AccessDenied</Code>
<Message>Access Denied</Message>
<RequestId>6A397XWG4ZQCPP2Y</RequestId>
<HostId>StIpx2AfO0kt9eZVe0XWWt8IisbcQdMfKbeYdbnJPAVgL6MamMNGhVvfqwxQ+EQ1zNL3xKgEmhI=</HostId>
</Error>


us-east-1.console.aws.amazon.com/s3/bucket/durva-feb/property/oo/edit?region=us-east-1
Amazon S3 > Buckets > durva-feb > Edit Object Ownership
Edit Object Ownership Info
Object Ownership
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.
Object Ownership
ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.
ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.
We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.
Enabling ACLs turns off the bucket owner enforced setting for Object Ownership
Once the bucket owner enforced setting is turned off, access control lists (ACLs) and their associated permissions are restored. Access to objects that you do not own will be based on ACLs and not the bucket policy.
I acknowledge that ACLs will be restored.
Object Ownership
Bucket owner preferred
If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.


us-east-1.console.aws.amazon.com/s3/buckets/durva-feb?region=us-east-1&tab=permissions
Amazon S3 > Buckets > durva-feb
Successfully edited Object Ownership.
durva-feb Info
Objects
Metadata
Properties
Permissions
Metrics
Management
Access Points
Permissions overview
Access finding
Access findings are provided by IAM external access analyzers. Learn more about How IAM analyzer findings work
View analyzer for us-east-1
Block public access (bucket settings)
Edit
Block all public access
On
Individual Block Public Access settings for this bucket


us-east-1.console.aws.amazon.com/s3/bucket/durva-feb/property/bpa/edit?region=us-east-1
Amazon S3 > Buckets > durva-feb > Edit Block public access (bucket settings)
Edit Block public access (bucket settings) Info
Block public access (bucket settings)
Block all public access
Block public access to buckets and objects granted through new access control lists (ACLs)
Block public access to buckets and objects granted through any access control lists (ACLs)
Block public access to buckets and objects granted through new public bucket or access point policies
Block public and cross-account access to buckets and objects through any public bucket or access point policies
Cancel
Save changes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Permissions
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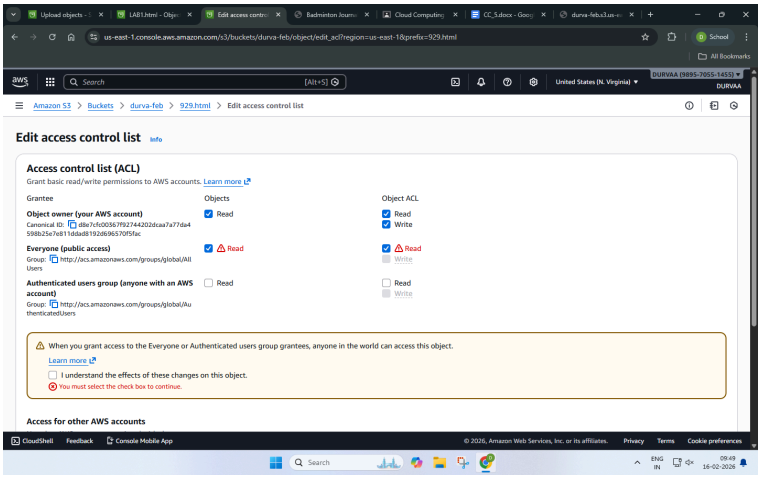
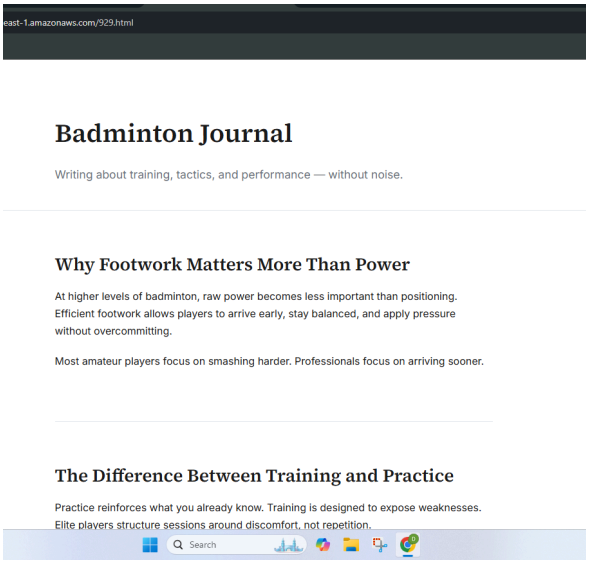

us-east-1.console.aws.amazon.com/s3/buckets/durva-feb/object/edit_acl?region=us-east-1&prefix=929.html
Amazon S3 > Buckets > durva-feb > 929.html > Edit access control list
Edit access control list
Access control list (ACL)
Grant basic read/write permissions to AWS accounts.
Grantee
Objects
Object ACL
Object owner (your AWS account)
Everyone (public access)
Authenticated users group (anyone with an AWS account)
Read
Write
When you grant access to the Everyone or Authenticated users group grantees, anyone in the world can access this object.
I understand the effects of these changes on this object.
You must select the check box to continue.
Access for other AWS accounts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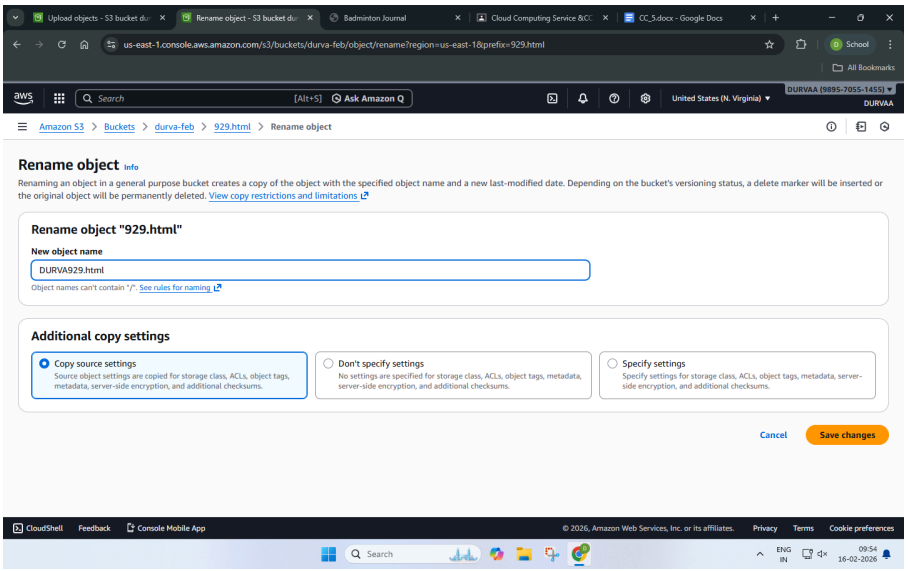
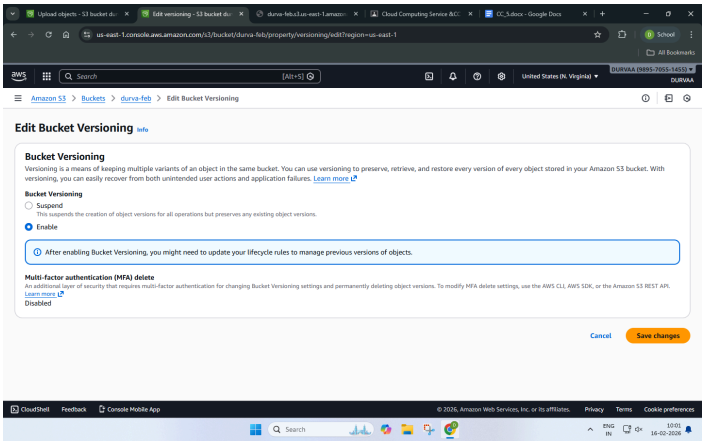
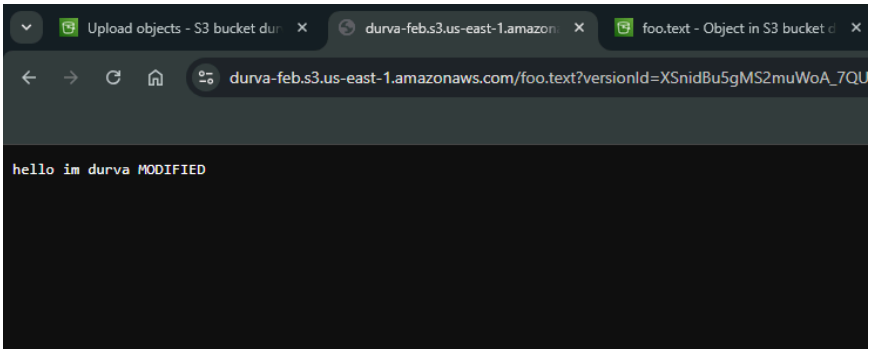
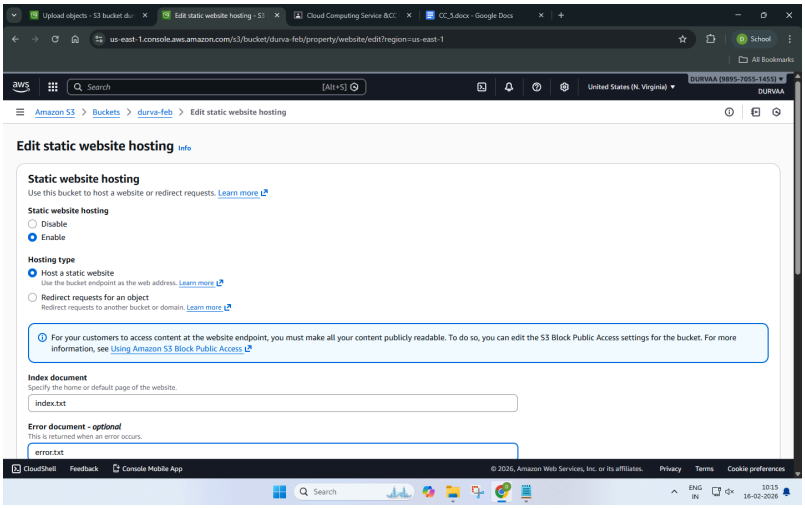
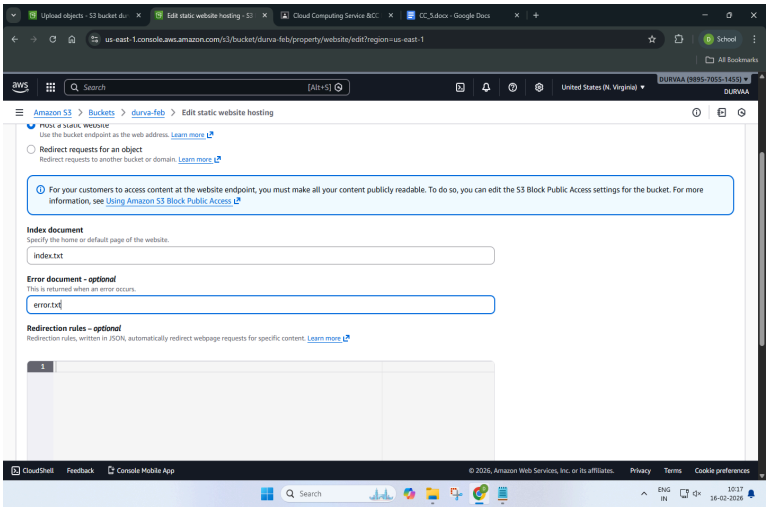

us-east-1.console.aws.amazon.com/s3/buckets/durva-feb/object/rename?region=us-east-1&prefix=929.html
Amazon S3 > Buckets > durva-feb > 929.html > Rename object
Rename object
Renaming an object in a general purpose bucket creates a copy of the object with the specified object name and a new last-modified date. Depending on the bucket's versioning status, a delete marker will be inserted or the original object will be permanently deleted.
Rename object "929.html"
New object name
DURVA929.html
Additional copy settings
Copy source settings
Source object settings are copied for storage class, ACLs, object tags, metadata, server-side encryption, and additional checksums.
Don't specify settings
No settings are specified for storage class, ACLs, object tags, metadata, server-side encryption, and additional checksums.
Specify settings
Specify settings for storage class, ACLs, object tags, metadata, server-side encryption, and additional checksums.
Cancel
Save changes

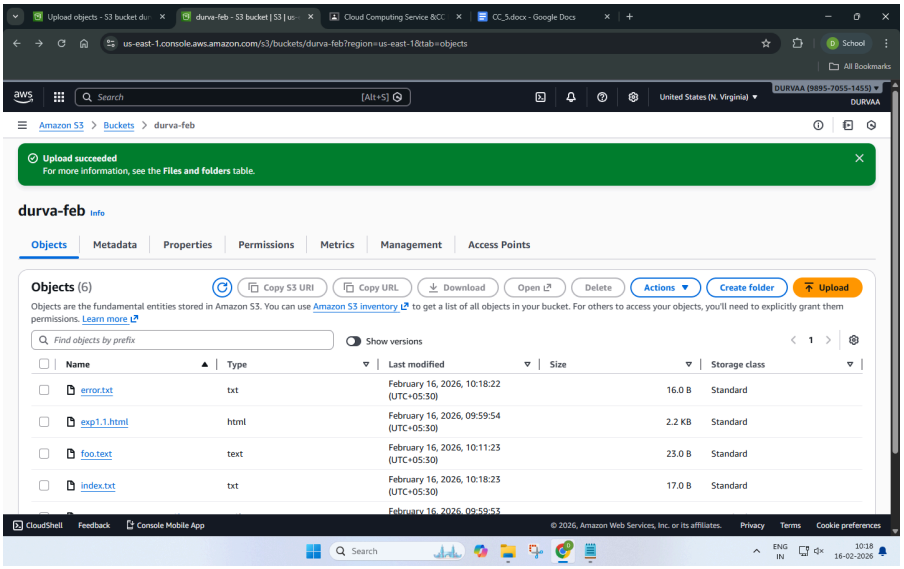

us-east-1.console.aws.amazon.com/s3/bucket/durva-feb/property/versioning/edit?region=us-east-1
Amazon S3 > Buckets > durva-feb > Edit Bucket Versioning
Edit Bucket Versioning
Bucket Versioning
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures.
Bucket Versioning
Suspend
This suspends the creation of object versions for all operations but preserves any existing object versions.
Enable
After enabling Bucket Versioning, you might need to update your lifecycle rules to manage previous versions of objects.
Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API.
Disabled
Cancel
Save changes

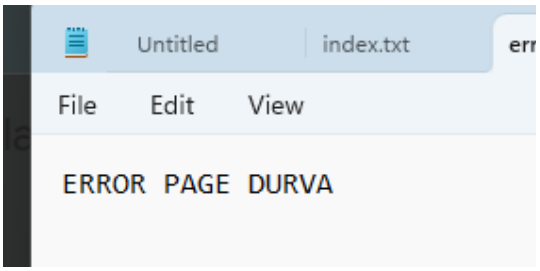

Upload objects - S3 bucket dur
durva-feb.s3.us-east-1.amazon
foo.text - Object in S3 bucket
durva-feb.s3.us-east-1.amazonaws.com/foo.text?versionId=XSnidBu5gMS2muWoA_7QU
hello im durva MODIFIED

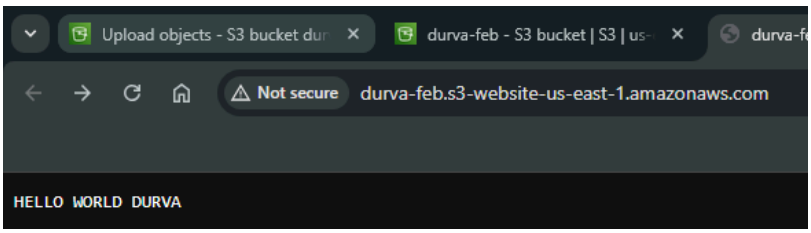
POST-EXPERIMENT EXERCISE:

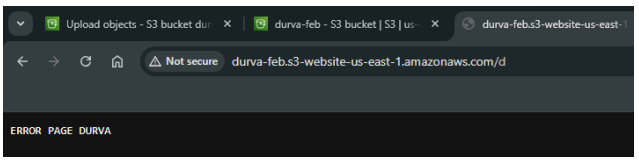


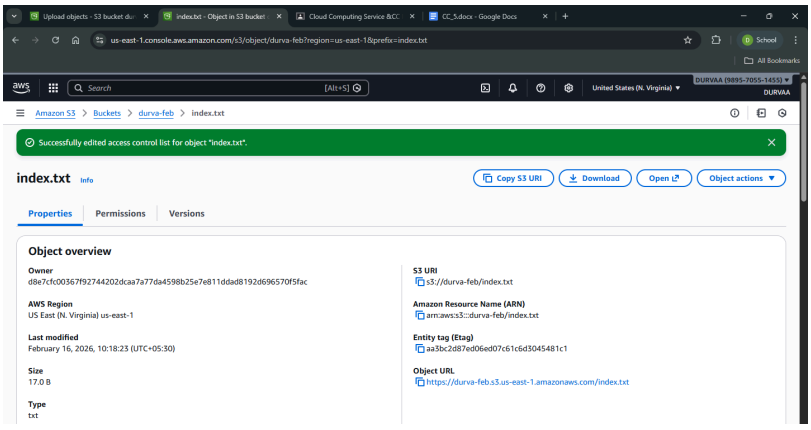


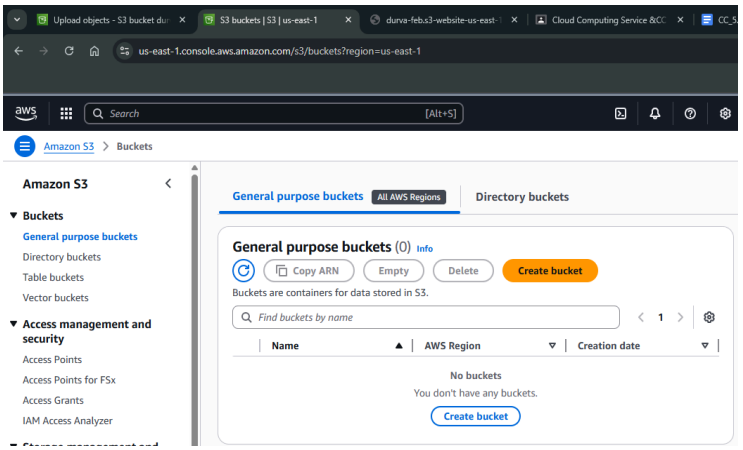












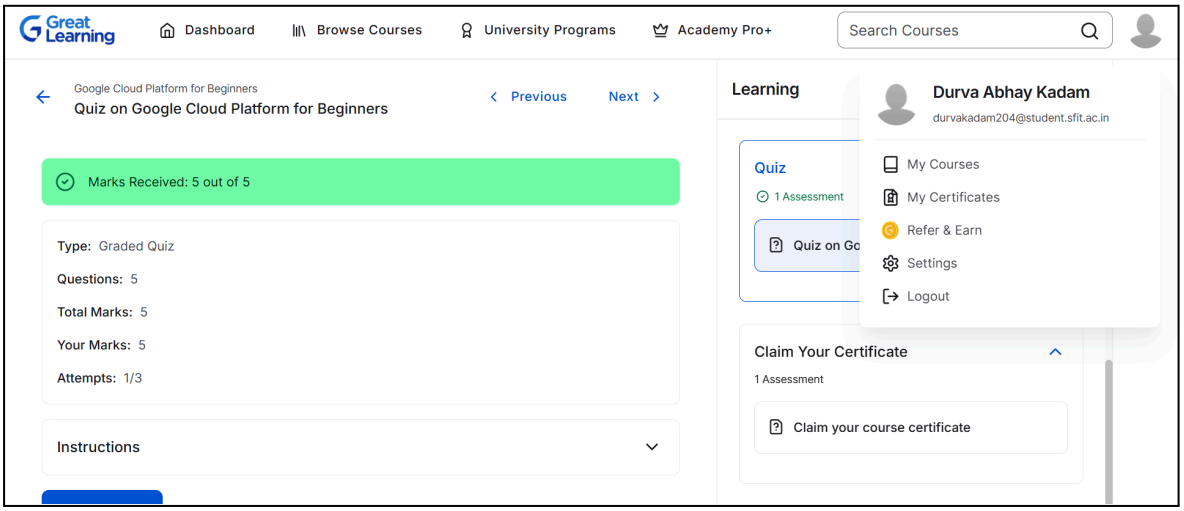
Extended Theory:

1. **Block level storage vs Object level storage(soft copy)**

ANS:

Block Level Storage	Object Level Storage
Stores data in fixed-size blocks	Stores data as objects (files + metadata)
Used in systems like SAN (Storage Area Network)	Used in cloud storage systems
Data is accessed by block address	Data is accessed using unique object IDs
Faster for databases and transactional systems	Better for large-scale unstructured data
Requires a file system to organize data	Built-in metadata management
Low latency and high performance	Highly scalable for huge data volumes
Harder to scale across distributed systems	Easily scalable across distributed systems
Examples: Amazon EBS, VMware VMFS	Examples: Amazon S3, OpenStack Swift

Post Experimental Exercise-

Questions:

1. A certificate course on GCP (1Hr)
https://www.mygreatlearning.com/academy/learn-for-free/courses/google-cloud-platform-for-beginners1?utm_source=chatbot&utm_medium=new_flow&utm_campaign=job
Submit test score and certificate of the attended course